Duality Synthesis in Quaternionic Logic: How Opposites Generate Truth

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Abstract

The quaternionic G-lattice defines logical connectives over $V = [0,1] \times [-1,1]^3$ with an SU(2) group action that separates truth (Re) from epistemic character (Im). The companion paper establishes that the imaginary dimensions are operationally distinguishable. This paper shows they are more: through the synthesis of opposites, they *generate* truth.

We define a synthesis operation on the prime algebra: given dual concepts a and $\bar{a}$ with prime signatures $\Phi(a)$ and $\Phi(\bar{a})$, their synthesis $a \oplus \bar{a} = \mathrm{lcm}(\Phi(a), \Phi(\bar{a}))$ produces a compound concept whose constructive complexity strictly exceeds that of either pole. The shared structure $\gcd(\Phi(a), \Phi(\bar{a}))$ forms an algebraic Markov blanket whose interiors are coprime by the Fundamental Theorem of Arithmetic, yielding a unique partition that resolves the observer-dependence critique of statistical blankets.

We introduce five axioms (C1–C5) for *constructive complexity measures* and show that both the prime factor count ω and the assembly index MA are instances. The gap operation—extracting what one concept has that another lacks—is proven to be irreducibly second-order (superinformation). Three algebraic tasks (directional subsumption, guaranteed composition, asymmetric gap analysis) are shown to be possible in the prime medium and impossible under cosine similarity.

The results establish that duality, operating through the imaginary dimensions of the G-lattice, is a generative mechanism: opposites do not cancel but produce new truth at higher algebraic layers.

1 Introduction

The prime-algebraic framework encodes semantic concepts as products of prime numbers [Ornelas Brand, 2026a], organises them into six algebraic layers [Ornelas Brand, 2026b], and identifies 14+ candidate dualities across those layers. The companion paper [Ornelas Brand, 2026c] constructs a quaternionic operator $\mathcal{O} = \{0, 1, +, i, j, k\}$ from which all six layers are recoverable as dimensional projections, and equips the resulting four-dimensional space with a G-lattice: a residuated lattice with an SU(2) group action whose commutation theorem characterises Boolean logic as the context-free special case.

That paper establishes that the imaginary dimensions (i, j, k) are operationally distinguishable from truth (r): agents at equal truth but different epistemic profiles make different decisions. Yet the G-lattice's tautologies depend only on Re; the imaginary dimensions enrich the *structure around* tautologies but do not alter which formulas are valid.

This paper addresses the question: *do the imaginary dimensions produce truth?* We show that they do, through the synthesis of opposites. When two dual concepts interact, their *relationship*—encoded in the imaginary dimensions—generates a compound concept with new crystallised content (Re) that neither pole possessed individually.

Contributions.

1. A synthesis operation on the prime algebra (Section 3) with a proof that it strictly increases constructive complexity (Theorem 5).
2. An algebraic Markov blanket theorem: the shared structure between any two concepts partitions uniquely into blanket and coprime interiors, by FTA (Section 4).
3. Five axioms for constructive complexity measures, with ω and MA as instances (Section 5).
4. A proof that the gap operation constitutes superinformation: irreducibly second-order (Section 6).
5. Three constructor-theoretic impossibility results distinguishing the prime medium from cosine similarity (Section 7).
6. Computational verification on all 14 duality pairs from the prime-algebraic ontology, with precise bilateral characterisation of which axes increase (Section 9).
7. A proof that the prime algebra is irreducible for synthesis: no function on quaternionic coordinates alone can compute lcm (Proposition 8).
8. A complexity–constructor bridge theorem: for media with unique decomposition, CCM axioms and constructor-theory conditions I1–I4 are equivalent; superinformation (I5) is the irreducible gap between the frameworks (Theorem 29).

2 Preliminaries

2.1 The prime algebra

Each semantic concept C is encoded as a prime signature $\Phi(C) = \prod_i p_i^{s_i(C)}$, where p_i is the i-th prime and $s_i(C) \in \{0, 1\}$ indicates whether primitive i is active for C [Ornelas Brand, 2026a]. By the Fundamental Theorem of Arithmetic (FTA), this encoding is injective: distinct concepts have distinct signatures.

Three operations are defined on signatures:

- $\gcd(\Phi(A), \Phi(B))$: shared structure (what A and B have in common).
- $\mathrm{lcm}(\Phi(A), \Phi(B))$: combined structure (the smallest concept containing both).
- $\Phi(A) \bmod \Phi(B)$: subsumption test ($= 0$ iff B divides A).

2.2 The G-lattice

The quaternionic G-lattice [Ornelas Brand, 2026c] is a residuated lattice on $V = [0, 1] \times [-1, 1]^3$ equipped with an SU(2) conjugation action. The real component r encodes crystallised truth; the imaginary triple (i, j, k) encodes epistemic character (potentiality, verifiability, recursive selection). The commutation theorem states that lattice connectives commute with the group action if and only if they depend exclusively on r; Boolean logic is the context-free special case.

2.3 Dualities

Definition 1 (Non-degenerate pair)**.** *A pair of concepts* (A, B) *is* non-degenerate *if neither* $\Phi(A) \mid \Phi(B)$ *nor* $\Phi(B) \mid \Phi(A)$*—equivalently, if both interiors* $\iota_A = \Phi(A)/\gcd(\Phi(A), \Phi(B)) > 1$ *and* $\iota_B = \Phi(B)/\gcd(\Phi(A), \Phi(B)) > 1$.

All theorems in this paper apply to arbitrary non-degenerate pairs: the synthesis, coprime-interiors, and complexity-increase results require only that both interiors be non-trivial. The 14 candidate dualities identified in Ornelas Brand [2026b] are a semantically motivated *subset* of non-degenerate pairs: they are pairs whose opposition is grounded in the layer structure (e.g., *creación/destrucción*, *orden/caos*). Eleven of the 14 are non-degenerate; three (*vida/muerte*, *consciente/ausente*, *individual/colectivo*) turn out to be degenerate under dependency expansion—one pole's primitives are a subset of the other's—so synthesis reduces to the more complex pole with no strict increase on any axis.

Remark 2 (Degenerate pairs reflect ontological dependency)**.** *The three degenerate pairs arise from intentional dependency-DAG design in Ornelas Brand [2026b]:* muerte *depends on* vida *(adding destrucción, caos, decaimiento, and muerte itself—19 vs. 15 primitives);* ausente *depends on* consciente *(adding pérdida, vacío, destrucción—22 vs. 18);* colectivo *depends on* individual *(adding muchos, cooperación, unión—20 vs. 16). In each case the second pole* presupposes *the first—death requires life, absence requires presence, collectivity requires individuality—so* $\Phi(\textit{first}) \mid \Phi(\textit{second})$ *by construction. This is a feature of the ontological design: these dualities express* existential dependency*, not algebraic opposition. The synthesis* lcm *reduces to the more complex pole because the "opposition" is already resolved within the dependency structure.*

3 The Synthesis Operation

Definition 3 (Synthesis of duals)**.** *Given a duality* $(a, \bar{a})$ *with prime signatures* $\Phi(a)$ *and* $\Phi(\bar{a})$*, the* synthesis *is:*

$$a \oplus \bar{a} \;=\; \mathrm{lcm}(\Phi(a),\ \Phi(\bar{a})) \tag{1}$$

The synthesis decomposes into three components:

$$\textit{Blanket:} \quad b = \gcd(\Phi(a),\ \Phi(\bar{a})) \tag{2}$$

$$\textit{Interior of } a: \quad \iota_a = \Phi(a) \,/\, b \tag{3}$$

$$\textit{Interior of } \bar{a}: \quad \iota_{\bar{a}} = \Phi(\bar{a}) \,/\, b \tag{4}$$

$$\textit{Synthesis:} \quad a \oplus \bar{a} = b \cdot \iota_a \cdot \iota_{\bar{a}} \tag{5}$$

Example 4 (Order from opposites)**.** *In the 72-primitive ontology of Ornelas Brand [2026b],* orden *and* caos *each depend on 7 primitives (including shared ones:* uno, fuerza, eje_profundidad, más, posición_temporal, mover*). The unique primitive of* orden *is* p_{83}*; that of* caos *is* p_{89}*. Then:*

$$\begin{aligned} b &= \gcd(\Phi(\textit{orden}),\ \Phi(\textit{caos})) && \textit{(6 shared primitives)} \\ \iota_{\textit{orden}} &= p_{83} && \textit{(the ordering direction)} \\ \iota_{\textit{caos}} &= p_{89} && \textit{(the disordering direction)} \\ \textit{orden} \oplus \textit{caos} &= b \cdot p_{83} \cdot p_{89} && (\omega = 6 + 1 + 1 = 8 > 7) \end{aligned}$$

The synthesis is not orden *AND* caos *(that would be* gcd $= b$*, the 6 shared primitives). It is not* orden *OR* caos *in the lattice sense (component-wise max of quaternionic coordinates). It is the concept that* contains both poles and their distinction*—a new entity at a higher level of constructive complexity. Since both unique primitives are in Layer 3 (J-axis), the bilateral support theorem predicts strict increase on* j *only, which is confirmed computationally.*

Theorem 5 (Synthesis increases complexity)**.** *Let* $(a, \bar{a})$ *be a duality with* $\Phi(a) \neq \Phi(\bar{a})$*. If each pole has at least one prime not shared with the other (i.e.,* $\iota_a > 1$ *and* $\iota_{\bar{a}} > 1$*), then:*

$$\omega(a \oplus \bar{a}) > \max(\omega(a),\ \omega(\bar{a})) \tag{6}$$

where $\omega(n)$ *denotes the number of distinct prime factors of* n*.*

Proof. By the inclusion-exclusion principle for prime factors:

$$\omega(\mathrm{lcm}(A, B)) = \omega(A) + \omega(B) - \omega(\gcd(A, B))$$

Let $s = \omega(\gcd)$, $\alpha = \omega(\iota_a) \geq 1$, and $\beta = \omega(\iota_{\bar{a}}) \geq 1$. Then $\omega(A) = s + \alpha$, $\omega(B) = s + \beta$, and:

$$\omega(\mathrm{lcm}) = (s + \alpha) + (s + \beta) - s = s + \alpha + \beta$$

Since $\alpha \geq 1$ and $\beta \geq 1$:

$$s + \alpha + \beta \;\geq\; s + 1 + \beta \;>\; s + \beta = \omega(B)$$

and symmetrically $s + \alpha + \beta > s + \alpha = \omega(A)$. Therefore $\omega(\mathrm{lcm}) > \max(\omega(A), \omega(B))$. □

Remark 6 (Synthesis is not a lattice connective)**.** *The synthesis $\oplus$ operates on prime signatures, not on quaternionic coordinates. It is algebraically distinct from the G-lattice connectives AND (= min) and OR (= max), which operate component-wise on V. The G-lattice connectives preserve or reduce complexity (min and max do not increase the number of active components); synthesis* increases *it. This is the formal sense in which opposites* generate *rather than* select.

Remark 7 (The synthesis in quaternionic coordinates)**.** *Theorem 5 operates on prime factor counts. The Synthesis Embedding Theorem of Ornelas Brand [2026c] lifts this to quaternionic coordinates: a layer-based projection $e\colon \{0,1\}^{72} \to [0,1]^4$ sends each concept's exponent vector to (r, i, j, k), and satisfies*

$$e(\mathrm{lcm})_X > \max(e(A)_X,\, e(B)_X) \iff \mathrm{Int}(A \,|\, B) \cap X \neq \emptyset \text{ and } \mathrm{Int}(B \,|\, A) \cap X \neq \emptyset$$

for each axis $X \in \{R, I, J, K\}$, where $\mathrm{Int}(A \mid B)$ is the set of primitives active in A but not in B. The strict increase on axis X thus occurs precisely when both interiors have X-support: primitives in the layer sets that contribute to X. This makes the "synthesis generates truth" claim a theorem with an exact characterisation, not a qualitative observation.

Proposition 8 (Irreducibility of the prime algebra)**.** *No function $f\colon [0,1]^4 \times [0,1]^4 \to [0,1]^4$ computes synthesis from quaternionic coordinates: there exist concepts A, B, C with $e(A) = e(B)$ but $e(\mathrm{lcm}(A, C)) \neq e(\mathrm{lcm}(B, C))$.*

Proof. The embedding e counts activated primitives per axis without distinguishing *which* primitives are active. Consider two concepts A (activating *eje_ vertical* [Layer 4] and *mover* [Layer 3]) and B (activating *eje_ lateral* [Layer 4] and *posición_ temporal* [Layer 3]). Each has exactly one Layer-4 and one Layer-3 primitive, so

$$e(A) = e(B) = \left(\tfrac{1}{36},\ \tfrac{1}{25},\ \tfrac{1}{40},\ 0\right).$$

Let C activate only *eje_ vertical*. Then $\Phi(C) \mid \Phi(A)$, so $\mathrm{lcm}(\Phi(A), \Phi(C)) = \Phi(A)$ and $e(\mathrm{lcm}(A, C)) = e(A)$. But $\Phi(C) \nmid \Phi(B)$, so $\mathrm{lcm}(\Phi(B), \Phi(C))$ adds a Layer-4 primitive to B, giving

$$e(\mathrm{lcm}(B, C)) = \left(\tfrac{2}{36},\ \tfrac{2}{25},\ \tfrac{1}{40},\ 0\right) \;\neq\; e(A).$$

If f existed, $e(A) = e(B)$ would force $f(e(A), e(C)) = f(e(B), e(C))$, contradicting $e(\mathrm{lcm}(A, C)) \neq e(\mathrm{lcm}(B, C))$. □

Remark 9. *The irreducibility is a feature of the architecture: the embedding captures the* structural *aspect of synthesis (which axes increase, by Remark 7), while the prime algebra retains the* identity *aspect (which specific primitives). The two representations are complementary—the prime algebra is irreducible for computing synthesis, and the G-lattice is irreducible for characterising its geometric structure.*

4 Algebraic Markov Blankets

The blanket $b = \gcd(\Phi(a), \Phi(\bar{a}))$ from the synthesis decomposition has a structural property that resolves a standing critique of statistical Markov blankets.

Theorem 10 (Coprime interiors)**.** *For any two concepts C_1, C_2 with prime signatures $\Phi(C_1), \Phi(C_2)$, define:*

- *Blanket:* $b = \gcd(\Phi(C_1), \Phi(C_2))$
- *Interior of* C_1*:* $\iota_1 = \Phi(C_1)/b$
- *Interior of* C_2*:* $\iota_2 = \Phi(C_2)/b$

Then $\gcd(\iota_1, \iota_2) = 1$.

Proof. By FTA, write $\Phi(C_j) = \prod_i p_i^{a_i^{(j)}}$. Then:

$$b = \prod_i p_i^{\min(a_i^{(1)}, a_i^{(2)})}, \qquad \iota_j = \prod_i p_i^{a_i^{(j)} - \min(a_i^{(1)}, a_i^{(2)})} \tag{7}$$

For each prime p_i, let $e_i = a_i^{(1)} - \min(a_i^{(1)}, a_i^{(2)})$ and $f_i = a_i^{(2)} - \min(a_i^{(1)}, a_i^{(2)})$.

If $a_i^{(1)} \leq a_i^{(2)}$, then $e_i = 0$, so $p_i \nmid \iota_1$. If $a_i^{(1)} > a_i^{(2)}$, then $f_i = 0$, so $p_i \nmid \iota_2$. In both cases $\min(e_i, f_i) = 0$. Therefore:

$$\gcd(\iota_1, \iota_2) = \prod_i p_i^{\min(e_i, f_i)} = 1$$

□

Remark 11 (Algebraic Markov property)**.** *The condition* $\gcd(\iota_1, \iota_2) = 1$ *is the algebraic analogue of conditional independence* $\iota_1 \perp \iota_2 \mid b$*: once the shared structure (blanket) is factored out, the remainders (interiors) share* nothing*. No prime factor "leaks" from one interior to the other through the blanket.*

Corollary 12 (Unique partition)**.** *The partition* (ι_1, b, ι_2) *is unique for any pair* (C_1, C_2)*.*

Proof. By FTA, $\Phi(C_1)$ and $\Phi(C_2)$ have unique prime factorisations. Hence $b = \gcd(\Phi(C_1), \Phi(C_2))$ and $\iota_j = \Phi(C_j)/b$ are each uniquely determined. □

Remark 13 (Relation to statistical Markov blankets)**.** *The prime-algebraic blanket addresses the* uniqueness *component of the observer-dependence critique [Bruineberg et al., 2018]: FTA guarantees a single partition, discovered by factorisation rather than chosen by an observer. The conditional-independence analogy (*$\gcd(\iota_1, \iota_2) = 1$ *as "no shared prime leaks through the blanket") is structural, not probabilistic: the prime algebra has no probability measure, so conditional independence in the statistical sense does not apply. What is gained is an exact, observer-free partition; what is not claimed is a reduction to the probabilistic framework.*

Proposition 14 (Nested blankets)**.** *If* $\Phi(A) \mid \Phi(B) \mid \Phi(C)$ *(divisibility chain), then at each level the Markov property holds: the interior of* A *relative to* B *and the interior of* B *relative to* C *are coprime.*

Proof. Apply Theorem 10 at each level of the chain. The FTA argument is identical at each scale. □

5 Constructive Complexity Measures

The synthesis operation increases the prime factor count ω. We now show that ω belongs to a general class of measures that detect constructive depth.

Definition 15 (Constructive complexity measure)**.** *A* ***constructive complexity measure*** *over a domain* D *with building blocks* B *and composition* $\oplus$ *is a function* $\mu : D \to \mathbb{Z}_{\geq 0}$ *satisfying:*

(C1) $\mu(b) = 0$ *for all* $b \in B$ *(grounding)*
(C2) $\mu(x) \geq 0$ *for all* $x \in D$ *(non-negativity)*
(C3) $\mu(a \oplus b) \leq \mu(a) + \mu(b) + c$ *for constant* $c \geq 0$ *(subadditivity)*
(C4) μ *is computable* *(computability)*
(C5) For all $a \in D$ *and* $b \in B$ *with* $b \nmid a$*:* $\mu(a \oplus b) > \mu(a)$ *(strict monotonicity)*

Remark 16. *C5 excludes degenerate measures (e.g.,* $\mu \equiv 0$*) and ensures that adding genuinely new building blocks always increases complexity.*

5.1 Instance 1: Assembly index

Walker & Cronin's assembly index MA [Sharma et al., 2023] operates on molecular structures with chemical joining as composition and $c = 1$. It counts the minimum number of joining operations needed to construct an object from basic fragments. High MA combined with high copy number is proposed as a substrate-independent signature of selection.

5.2 Instance 2: Prime factor count

The join count $\mu(x) = \omega(x) - 1$ (for $x > 1$; $\mu(1) = 0$) operates on semantic concepts with lcm as composition and $c = 1$. Here ω is the number of distinct prime factors. C1–C5 are immediate: grounding ($\omega(p_i) = 1$, so $\mu = 0$), non-negativity, subadditivity (by inclusion-exclusion), computability (by trial division), and strict monotonicity (adding a new prime increases ω by 1).

5.3 Structural parallel

Proposition 17 (Structural parallel)**.** *MA and ω share three properties beyond C1–C5:*

1. ***Overlap correction:*** $\omega(\mathrm{lcm}(A, B)) = \omega(A) + \omega(B) - \omega(\gcd(A, B))$*; for MA, shared subassemblies reduce the combined index analogously.*
2. ***Finite irreducible basis:*** *primes (resp. basic molecular fragments).*
3. ***Discrimination of structured from random:*** *the Domain Validity Score [Ornelas Brand, 2026b] perfectly discriminates 8 real domains from 3 pseudoscience controls (gap* 0.233*, zero errors).*

They diverge on one key point: ω has ***unique decomposition*** *(FTA), while MA does not—multiple assembly pathways may yield the same index. This makes ω strictly stronger as an algebraic invariant.*

Proposition 18 (Rarity of high complexity under random assembly)**.** *In a prime product space with k building blocks, under uniform random selection from all 2^k possible objects, the probability of obtaining an object with complexity $\mu \geq m$ is the upper tail of* Binomial$(k, 1/2)$*:*

$$P(\mu \geq m) = \frac{1}{2^k} \sum_{j=m+1}^{k} \binom{k}{j} \tag{8}$$

For $m \gg k/2$, this decays exponentially: $P(\mu \geq m) \leq \exp(-2(m - k/2)^2/k)$ by Hoeffding's inequality.

Proof. The number of objects with exactly j prime factors is $\binom{k}{j}$. Under uniform selection, $P(\omega = j) = \binom{k}{j}/2^k = \mathrm{Binomial}(k, 1/2)$. Since $\mu = \omega - 1$, $P(\mu \geq m) = P(\omega \geq m + 1)$. The exponential bound follows from Hoeffding's inequality. □

Conjecture 19 (Universal Selection Signature)**.** *For any constructive complexity measure μ satisfying C1–C5 over a domain D with k building blocks: the empirical distribution of μ values in a population produced by selection is stochastically dominated by the Binomial$(k, 1/2)$ null of Proposition 18, with separation detectable by a one-sided Kolmogorov–Smirnov test at sample sizes polynomial in k.*

Remark 20 (Testability)**.** *The USS is not straightforwardly testable on the prime-algebraic ontology: the 72 primitives* are *the building blocks, so their ω values (*1–22*, mean* 9.4*) are low by construction, not by selection. The relevant null (uniform over 2^{72} products, mean $\omega = 36$) applies to* compound concepts*, not to the basis. The closest existing empirical test is the Domain Validity Score discrimination of Ornelas Brand [2026b] (gap* 0.233*, zero misclassifications across 11 domains), which confirms that ω-derived measures detect structure in real data. A direct USS test requires a corpus of naturally-occurring compound concepts with known prime signatures.*

6 The Prime Algebra as Information Medium

We verify that the prime algebra satisfies the five conditions for an information medium under Deutsch & Marletto's Constructor Theory of Information [Deutsch & Marletto, 2015].

Proposition 21 (Conditions I1–I4)**.** *The prime algebra satisfies:*

- ***I1 Distinguishability:*** *For distinct $C_1 \neq C_2$, $\Phi(C_1) \neq \Phi(C_2)$ by construction. The factorisation algorithm recovers the unique decomposition.*
- ***I2 Clonability:*** *Integer assignment copies any signature exactly to a blank substrate (memory initialised to 1).*
- ***I3 Non-triviality:*** *With k primes, $2^k \geq 2$ distinguishable states.*
- ***I4 Interoperability:*** *Measurement (factorisation), copying (assignment), and computation (*mod, lcm, gcd*) are all exact, deterministic, and composable within the same integer medium.*

6.1 Superinformation (Condition I5)

Proposition 22 (Condition I5 satisfied)**.** *The gap operation* $\mathrm{gap}(A, B) = \Phi(A)/\gcd(\Phi(A), \Phi(B))$ *constitutes superinformation in the prime-algebraic medium.*

Proof. Three requirements.

(i) The gap is second-order. A first-order information task maps a single variable to an output: $f : \Phi(A) \mapsto \Phi(A')$. The gap requires *two* variables as input and outputs a *relationship*—the structural difference between two concepts—not the content of either alone.

(ii) The gap is irreducible to first-order. Suppose a function $g : \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ exists with $\mathrm{gap}(A, B) = g(\Phi(A))$ for all B. Fix A with $\Phi(A) = p_1p_2p_3$ and choose B_1 with $\Phi(B_1) = p_1$, B_2 with $\Phi(B_2) = p_1p_2$. Then $\mathrm{gap}(A, B_1) = p_2p_3 \neq p_3 = \mathrm{gap}(A, B_2)$, but $g(\Phi(A))$ is a fixed value. Contradiction.

(iii) The output remains in the medium. Since $\gcd(\Phi(A), \Phi(B))$ divides $\Phi(A)$, the quotient is a positive integer whose prime factors are a subset of those of $\Phi(A)$. □

Remark 23 (Asymmetry)**.** *In general* $\mathrm{gap}(A, B) \neq \mathrm{gap}(B, A)$*: the gap encodes an* oriented *relationship. This directional structure is characteristic of superinformation.*

Remark 24 (Gap as the engine of synthesis)**.** *The gap is exactly the interior from Definition 3:* $\mathrm{gap}(a, \bar{a}) = \iota_a$ *and* $\mathrm{gap}(\bar{a}, a) = \iota_{\bar{a}}$*. The superinformation proof therefore shows that the* distinguishing content *of each pole—the raw material of synthesis—is irreducibly second-order. Synthesis is not reducible to operations on either pole alone; it requires the relationship between them.*

7 The Possible/Impossible Distinction

The prime algebra provides a concrete instance of Constructor Theory's core innovation: a distinction between possible and impossible tasks across information media.

Theorem 25 (Three impossibility results)**.** *The following three tasks are* ***possible*** *in the prime-algebraic medium and* ***impossible*** *under cosine similarity:*

Proof. **Task 1: Directional subsumption.** $A \supseteq B$ is directional: $A \supseteq B$ does not imply $B \supseteq A$.

- **Prime:** $\Phi(A) \bmod \Phi(B) = 0 \iff A \supseteq B$. Directional by construction. *Possible.*
- **Cosine:** $\cos(\vec{A}, \vec{B}) = \cos(\vec{B}, \vec{A})$ (inner product is symmetric). No function of cosine alone can distinguish "A contains B" from "B contains A." *Impossible.*

Task 2: Guaranteed composition. Produce C guaranteed to subsume both A and B.

- **Prime:** $C = \mathrm{lcm}(\Phi(A), \Phi(B))$. Algebraically guaranteed. *Possible.*
- **Cosine:** $\vec{C} = \vec{A} + \vec{B}$ does not guarantee $\cos(\vec{C}, \vec{A}) \geq \tau$ for any threshold (e.g., $\vec{A} = -\vec{B}$ gives $\vec{C} = \vec{0}$). *Impossible.*

Task 3: Asymmetric gap analysis. Extract what A has that B doesn't, and vice versa, as distinct objects.

- **Prime:** $\iota_A = \Phi(A)/\gcd(\Phi(A), \Phi(B))$, $\iota_B = \Phi(B)/\gcd(\Phi(A), \Phi(B))$, with $\gcd(\iota_A, \iota_B) = 1$ by FTA. *Possible.*
- **Cosine:** $\vec{A} - \vec{B} = -(\vec{B} - \vec{A})$ is antisymmetric but does not decompose into separable components unique to each side. *Impossible.*

□

Remark 26. *Task 1 (directional subsumption) is impossible under* any *symmetric similarity, not only cosine: the inner product $\langle \vec{A}, \vec{B} \rangle = \langle \vec{B}, \vec{A} \rangle$ cannot distinguish "A contains B" from "B contains A." Tasks 2 and 3 are impossible under cosine specifically—the standard metric in modern NLP (word2vec, BERT, sentence transformers)—but could in principle be recovered by richer vector-space operations (e.g., element-wise max for Task 2, signed residuals for Task 3). The practical force of the impossibility is that cosine is what practitioners use.*

Remark 27. *The three tasks map directly to the synthesis decomposition: Task 1 tests subsumption ($\Phi(a) \mid \Phi(c)$), Task 2 computes synthesis (*lcm*), and Task 3 extracts the interiors ($\iota_a, \iota_{\bar{a}}$). The impossibility results therefore show that the* synthesis of opposites *is impossible in flat vector spaces—it requires the Boolean lattice structure of the prime algebra.*

8 The Rule of Three as Constructor

The synthesis $a \oplus \bar{a} = \mathrm{lcm}(\Phi(a), \Phi(\bar{a}))$ combines two concepts. A related operation—analogical completion—requires three. The Rule of Three, $C_4 = (\Phi(C_2) \times \Phi(C_3))/\Phi(C_1)$, introduced in Ornelas Brand [2026a], solves $A : B = C :?$ deterministically when $\Phi(C_1) \mid \Phi(C_2) \times \Phi(C_3)$.

This satisfies Deutsch's definition of a **constructor** [Deutsch, 2013]: an entity that causes a specific transformation and retains the capacity to repeat it. It also constitutes a fourth impossibility under cosine similarity: the proportional relationship $\Phi(C_4)/\Phi(C_3) = \Phi(C_2)/\Phi(C_1)$ requires directed ratios, not symmetric dot products.

Remark 28 (Relation to synthesis)**.** *The Rule of Three and the synthesis operation are complementary constructors on the prime algebra. Synthesis combines* opposites *(*lcm*: what both have plus what distinguishes them). The Rule of Three completes* proportions *(multiplication and division: the structural transfer from one pair to another). Both are possible in the prime medium and impossible under cosine similarity; both are deterministic constructors in Deutsch's sense.*

9 Computational Verification

All results were verified on the 72 primitives and 14 duality pairs of Ornelas Brand [2026b]. The layer-based embedding of Ornelas Brand [2026c] maps each concept's dependency-expanded signature to $(r, i, j, k) \in [0, 1]^4$, with axis-contributing sets $|R| = 36$, $|I| = 25$, $|J| = 40$, $|K| = 4$.

Table 1: Verification of synthesis properties on the 14 duality pairs from Ornelas Brand [2026b].

Test	N	Violations	Result
$\omega(\text{lcm}) > \max(\omega(a), \omega(\bar{a}))$ for non-degenerate pairs	11	0	exact
Coprime interiors ($\gcd(\iota_a, \iota_{\bar{a}}) = 1$)	14	0	exact
$e(\text{lcm}) \geq \max(e(a), e(\bar{a}))$ componentwise	14	0	exact
Bilateral support $\Leftrightarrow$ strict increase	11	0	exact
Degenerate pairs detected ($\Phi(a) \mid \Phi(\bar{a})$)	3	0	exact
Divisibility preserves order (random pairs)	10 000	0	exact
lcm $\geq$ max (random pairs)	10 000	0	exact

Of the 11 non-degenerate duality pairs, 9 show strict increase on r (crystallised truth), and all 11 show strict increase on at least one axis. The two pairs without r-increase (*orden*/*caos*, *creación*/*destrucción*) have interiors only in Layer 3 (J-axis), correctly predicted by the bilateral support theorem. The two Layer 6 pairs (*temporal_obs*/*eterno_obs*, *receptivo*/*creador_obs*) show strict increase on all four axes, as their interiors span all contributing layer sets.

What distinguishes dualities from arbitrary pairs. Duality pairs have a mean Jaccard overlap of 0.614 (they share most of their primitives), compared to 0.076 for random pairs of the same sizes (Kolmogorov–Smirnov $p < 10^{-4}$). This high overlap is precisely what makes duality synthesis produce *small, targeted* gains: the mean complexity increase is 1.43 primes (vs. 5.97 for random pairs, which share almost nothing). The synthesis of opposites generates less raw complexity than combining unrelated concepts, but the gain is *semantically structured*: it fills the specific gap between the two poles, not arbitrary corners of the concept space.

10 Discussion

10.1 Measuring vs. enabling

Constructive complexity (Section 5) and constructor-theoretic information (Section 6) address complementary questions: *how much was built?* (backward-looking) and *what can be built?* (forward-looking). The prime algebra bridges both: it has a constructive complexity measure (ω) and satisfies constructor-theoretic conditions (I1–I5), suggesting that these are two faces of the same coin.

Theorem 29 (Complexity–constructor bridge)**.** *Let $(D, B, \oplus)$ be a domain with finite basis $B = \{b_1, \ldots, b_k\}$, a join-composition $\oplus$ satisfying $S(a \oplus b) = S(a) \cup S(b)$ for a support function $S\colon D \to 2^B$, and unique decomposition (every element has a unique support).*

(i) *$\mu(x) = |S(x)| - 1$ is a CCM satisfying C1–C5.*

(ii) *The medium satisfies I1–I4 of Constructor Theory.*

(iii) *I5 (superinformation) is independent: it holds iff the medium admits a second-order gap $\text{gap}(a, b)$ with $S(\text{gap}) = S(a) \setminus S(b)$.*

The prime algebra satisfies all three: FTA provides unique decomposition, lcm *is the join, and integer division provides the gap.*

Proof. (i) C1: $|S(b_i)| = 1$, so $\mu(b_i) = 0$. C2: $|S(x)| \geq 1$ for all x, so $\mu \geq 0$. C3: $|S(a) \cup S(b)| \leq |S(a)| + |S(b)|$, giving $\mu(a \oplus b) \leq \mu(a) + \mu(b) + 1$ with $c = 1$. C4: unique support over a finite basis is decidable (enumerate and check). C5: if $b \notin S(a)$, then $S(a \oplus b) = S(a) \cup \{b\}$, so $\mu(a \oplus b) = \mu(a) + 1 > \mu(a)$.

(ii) I1: distinct elements have distinct supports by uniqueness; the support serves as a distinguishing certificate. I2: copying the support to a blank substrate clones the element. I3: $|B| \geq 2$

gives ≥ 2 distinguishable states. I4: $\oplus$, support extraction, and equality testing are computable over a finite basis.

(iii) C1–C5 constrain a function $\mu\colon D \to \mathbb{Z}_{\geq 0}$ on individual elements. The gap takes two inputs and outputs a relationship; by Proposition 22, it is irreducibly second-order (varying the second argument changes the output even when the first is fixed). Therefore I5 is not entailed by C1–C5. Conversely, C1–C5 do not prohibit I5: the prime algebra satisfies both. □

10.2 Synthesis and the four axes

The layer-based embedding of Ornelas Brand [2026c] distributes the 72 primitives across four axis-contributing sets: R (Layers 1,2,4,6), I (Layers 4,6), J (Layers 3,5,6), K (Layer 6 only). This yields a precise connection between synthesis and the quaternionic axes.

The Bilateral Synthesis Theorem (Remark 7) states that axis X shows strict increase iff both interiors have X-support. Therefore:

- A duality whose interiors differ only in Layer 3 or 5 primitives (e.g., *orden*/*caos*, *creación*/*destrucción*) produces strict increase on j but not on r: the synthesis generates new *verifiable* structure.
- A duality whose interiors include Layer 4 primitives (e.g., *bien*/*mal*, *verdad*/*mentira*) produces strict increase on r and i: the synthesis generates new crystallised truth *and* potentiality.
- Layer 6 dualities (*temporal_obs*/*eterno_obs*, *receptivo*/*creador_obs*) produce strict increase on all four axes, including k: recursive selection is engaged only at the observer level.

This replaces the earlier informal mapping (blanket$\to r$, interiors$\to i$, etc.) with a precise, layer-determined characterisation: which axes increase depends on *where* in the dependency structure the interiors diverge, not on a fixed assignment of synthesis components to axes.

10.3 Relation to Lawvere's categorical synthesis

The synthesis $a \oplus \bar{a} = \mathrm{lcm}(\Phi(a), \Phi(\bar{a}))$ shares the pattern of Lawvere's formalisation of Hegelian *Aufhebung* [Lawvere, 1969, 1996]: thesis and antithesis produce a synthesis that preserves both. The precise categorical status is as follows.

The prime algebra, ordered by divisibility, is a distributive lattice and therefore a thin category. In this category, lcm is the *coproduct* (categorical join) and gcd is the *product* (categorical meet)—standard lattice theory. The coprime-interiors theorem (Theorem 10) adds content beyond the coproduct: it guarantees that the join decomposes uniquely into a shared blanket and disjoint contributions from each pole.

Lawvere's Aufhebung, however, operates on *levels* of a cohesive topos via adjoint functors between connected-component functors at different scales [Lawvere, 1996]. Our algebra has no topos structure, no connected-component functors, and no adjunction between levels. The parallel is therefore at the level of the *dialectical pattern*—opposition $\to$ preservation $\to$ elevation—not at the level of the categorical formalism. We share Lawvere's conclusion (synthesis produces genuinely new structure) but reach it by a different route (unique factorisation in a distributive lattice, not adjoint functors in a topos).

11 Conclusion

The imaginary dimensions of the quaternionic G-lattice are not decoration. Through the synthesis of opposites—formalised as lcm in the prime algebra—they generate compound concepts whose constructive complexity strictly exceeds that of either pole (Theorem 5). The shared structure between any two concepts partitions uniquely into an algebraic Markov blanket and coprime interiors (Theorem 10), resolving the observer-dependence critique. The gap operation—the raw

material of synthesis—is irreducibly second-order (Proposition 22), and three algebraic tasks that synthesis enables are provably impossible under cosine similarity (Theorem 25).

The Bilateral Synthesis Theorem of Ornelas Brand [2026c] lifts these results from prime factor counts to quaternionic coordinates, characterising exactly which axes show strict increase for each duality pair (Section 9). Of 14 tested dualities, the 11 non-degenerate pairs all confirm the theorem; the 3 degenerate pairs (one pole subsuming the other) correctly produce no increase. The prime algebra is shown to be irreducible for this purpose: no function on quaternionic coordinates alone can compute synthesis (Proposition 8)—the embedding captures geometric structure while the prime algebra retains combinatorial identity. The bridge between constructive complexity (C1–C5) and constructor-theoretic information (I1–I5) is made precise (Theorem 29): for media with unique decomposition the two frameworks are equivalent up to superinformation, which is the irreducible second-order condition.

The companion paper [Ornelas Brand, 2026c] showed that Boolean logic is what survives when all perspectives are averaged: the context-free special case. This paper shows what happens when perspectives *interact*: opposites, operating through the imaginary dimensions, produce new truth. Boolean logic captures the panorama; duality synthesis captures the engine.

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